INTERIM MILITARY SPECIFICATION

BINOCULAR, HAND-HELD, FOR MARINE USE

1. SCOPE

1.1 Scope. - This specification covers hand-held binoculars for Marine use.

1.2 Classification. - Binoculars shall be of the following types as specified (see 6.1):

Type I - 7-power, 50 millimeter (mm) - aperture.
Type II - 6-power, 42 mm - aperture.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
QQ-S-781 - Strapping, Flat; Steel.
UU-C-843 - Cushioning Materials, Cellulosic.
LLL-B-631 - Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY
JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116 - Preservation, Methods of.
JAN-P-139 - Packaging and Packing for Overseas Shipment - Plywood, Container Grade.
JAN-F-675 - Films, Reflection-Reducing, for Glass Optical Elements.
MIL-S-853 - Steel, Corrosion-Resisting: Bars and Forgings (Except for Reforging).
MIL-C-2910 - Covering, Organic-Plastic (for Instrument Bodies).

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

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DRAWINGS

BUREAU OF SHIPS

S2407-533150 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Assembly.
S2407-533151 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533152 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533153 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533154 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533155 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533156 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533157 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533158 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533159 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533160 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533161 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533162 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533163 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533164 - Binoculars, Prismatic, 7 x 50 Power, Mark 45, Mod. O, Details.
S2407-533198 - Standard 7 x 50 Binoculars Optical Assembly and Details.
S2407-533370 - 7 x 50 Binocular Mark 32, Mod. 7, General Assembly.
S2407-533371 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533372 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533373 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533374 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533375 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533376 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533377 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533378 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533379 - 7 x 50 Binocular Mark 32, Mod. 7, Details.
S2407-533380 - 7 x 50 Binocular Mark 32, Mod. 7, Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE
Consolidated Freight Classification - Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

UNITED STATES POST OFFICE
Postal Regulations.

(Application for copies should be addressed to the United States Post Office, Washington 25, D.C.)

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3. REQUIREMENTS

3.1 General. - Hand-held binoculars shall be of the prismatic stereoscopic type, wherein the image in each of the two telescopic systems is erected by a prism system designed to enhance stereoscopy.

3.2 Material. -

3.2.1 Metal. - The metal used shall be of the highest quality, and, where practicable, shall be selected from the stand-point of resistance to the corrosive and galvanic action of salt air, moisture, tropical climates, and extreme temperatures likely to be encountered in the Marine service. Particular attention shall be given to the quality of any castings used.

3.2.2 Glass. - The glass used for the optical elements shall be stable, nonhygroscopic, free from strains, striae, bubbles, seeds, or stones visible in the field of view or which would impair optical performance.

3.2.3 Lubricating and sealing materials. - All greases, waxes, sealing materials, and lubricants used on the metal parts shall be such as not to cause corrosive action on the optical and mechanical parts, cause filming of the optical elements, nor support fungus or mite growth.

3.3 Optical. -

3.3.1 The optical parts and the entire interior of the binoculars shall be clean and free from oil, dust, dirt, or other foreign matter and free from any lubricant which might, in time, collect on any of the optical parts. There shall be no dust specks on surfaces obvious to the unaided eye, when looking directly into the binocular, against a background having the brightness of the sky in average daylight.

3.3.2 Optical surfaces, except reflecting and cemented surfaces, shall be coated in accordance with Specification JAN-F-675.

3.3.3 The equivalent focal lengths of the two objectives and of the two eyepieces shall not differ by more than 1.5 percent. Objectives, mounted eyepieces, and eyepiece elements furnished for repair or maintainence parts shall not vary more than plus or minus 1 percent from the nominal focal lengths.

3.3.4 The image shall be sensibly free from defects resulting from spherical aberration and distortion. The image shall be sensibly free from defects resulting from coma and chromatic aberration within 1.5 degrees of true field about the center of the field and shall be reasonably free from these defects over the rest of the field. Aberrations which cannot be eliminated, and which affect either the definition or the stereoscopic accuracy of the instrument shall be substantially identical in both fields of view.

3.3.5 The images of a distant vertical straight line formed by the two optical systems shall be parallel to each other within 1 degree and neither image shall vary from the vertical by more than 1 degree. When two parallel pencils or bundles of rays are projected into the two objectives, the conjugate rays emerging from the two eyepieces shall be parallel within the following limits at the infinite focus of the eyepiece and for any setting of the interocular distance scale:

(a) Fourteen minutes of arc (apparent field) in a direction perpendicular to the plane determined by the two entrant pencils.
(b) They shall not diverge more than 28 minutes of arc (apparent field), nor converge more than 14 minutes of arc (apparent field), in the direction parallel to the plane of the two entrant pencils.

3.3.6 Ghost images or images of targets outside the field of view formed by extraneous reflections within the binocular, are not permissible when looking through the binocular with the eyes in a normal position.

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3.4 Reticle. - When specified (see 6.1), a reticle of a design approved by the bureau or agency concerned, shall be mounted in a fixed position in the focal plane of the objective on the left side of the binocular so that the horizontal meridian is approximately parallel to the line of centers of the left and right objectives for 63 mm interpupillary distance adjustment. Binoculars equipped with reticles shall read correctly within plus or minus 3 percent.

3.5 Mechanical. -

3.5.1 The eyepiece shall be of the rotating, focusing type. Each eyepiece shall be capable of being focused independently from plus 2 to minus 4 diopters. There shall be a knurled strip around the circumference of each eyepiece barrel to facilitate focusing. A diopter scale shall be provided for each eyepiece, with a line for the zero position and one for each diopter, every second diopter line to be numbered. The index shall be fixed and the scale shall be on the rotating part. The zero mark of the diopter scale shall be correctly located within 1/4 diopter, and the interval between adjacent lines shall be correct to 1/4 diopter. Less than one rotation of the eyepiece shall be required to change the focus through the entire range of 6 diopters. There shall be index lines on the underside of the binocular (adjacent to the staple) and on the fixed portion of the eyepiece tubes for use in setting the diopter scales.

3.5.2 Unless otherwise specified in the contract or order, each ocular shall be provided with a moulded eyeguard cap of such shape that any cross-section at right angles to the line of sight will be circular and concentric with the ocular axis. The size and shape of the caps shall be such as to allow the observer to see the entire field of view when the eyes are placed in the most comfortable position with respect to the two caps, and exclude from both eyes the major portion of all light except that transmitted by the ocular. The inner surfaces of the caps shall be corrugated or ridged so as to eliminate stray light, and the caps shall be readily removable to facilitate cleaning the exit surfaces of the eye lenses.

3.5.3 The hinges of the binoculars shall be designed to permit the adjustment of the interocular distance from 56 to 74 mm, with a scale graduated at 2 mm intervals. The interpupillary distance indicated by the scale shall be correct to within 1 mm. The hinge shall be fitted and adjusted with sufficient tightness and the hinge lubricant shall be of such a consistency that the interpupillary distance shall be maintained when the binoculars are tested as specified in 4.4.3. The hinge lugs shall be integral with the respective body tubes.

3.5.4 The main bodies shall be die castings, centrifugal castings, or extrusions of an aluminum base alloy.

3.5.5 There shall be a staple or loop, of suitable strength, integral with each main body tube, near the ocular end, for attaching the sling strap, and so positioned as to facilitate wearing the binoculars on a strap around the neck, and placed and balanced properly for ready use.

3.5.6 Cover plates shall be used at the ends of the main body tubes and shall overlap the tubes slightly. A gasket of suitable material shall be placed between the two parts if required to insure moisture tightness. The cover plates shall be secured by a sufficient number of screws, suitably located to insure that the cover plates will be held securely in place.

3.5.7 Screws shall be made of corrosion resistant steel and shall conform to class 7 of Specification MIL-S-853.

3.5.8 The objectives shall be secured in double eccentric mounts to permit adjustment in any direction in a plane perpendicular to the line of sight. A cover shall be provided to prevent tampering with the adjustment.

3.6 The binoculars shall be spraytight or watertight as specified (see 6.1). No water shall penetrate into the interior of the binoculars when subjected to the spraytight test specified in 4.4.4 or the watertight test specified in 4.4.5.

3.7 Shockproofness. - The binoculars shall withstand the shock test specified in 4.4.6 without damage or without throwing the collimation out of adjustment.

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3.8 Accessories.-

3.8.1 Each binocular shall be equipped with a black vinylite sling strap of a length sufficient to allow the binocular to rest upon the chest when the strap is placed around the neck.

3.8.2 Each binocular shall be supplied with a black vinylite carrying case as specified (see 6.1), in order that any special accessories such as filters or rubber eyeguards may be accommodated in the case.

3.9 Finish.-

3.9.1 The main bodies and all aluminum parts shall be anodized and dyed black before subsequent finishing. The main body tubes shall be covered with organic plastic in accordance with Specification MIL-C-2910. Exposed metal surfaces shall be covered with a durable black baked on lacquer having a suitable synthetic base.

3.9.2 The prisms shall not be silvered.

3.9.3 Finishes of the metal parts shall be such as not to permit corrosive action of the optical or mechanical parts.

3.9.4 Polished surfaces shall be free from objectionable scratches, pits, stains, and other defects visible in the field of view or that would impair optical performance.

3.9.5 No bright reflections shall be visible when the exit pupil is viewed from a point at some distance along the ocular axis extended. All surfaces at which troublesome reflections might occur shall be blackened by a suitable process. Any material or process employed for this purpose shall be such that it will not cause dusting or flaking.

3.10 Marking.- Each binocular shall be marked on the left cover plate with the nominal power, aperture, and field. Additional marking to identify using activity, contractor, serial number, etc., shall be specified (see 6.1).

3.11 Type I - 7 power, 50 mm - aperture.-

3.11.1 The magnification of each of the two telescopic systems of the binocular shall be seven diameters plus or minus 2 percent. The magnification of the two telescopes of a single binocular shall not differ by more than 1 percent.

3.11.2 The effective aperture of the objective shall be not less than 50.0 mm, nor more than 52.5 mm.

3.11.3 The central exit pupil shall have a diameter not less than 50 mm divided by the measured magnification except that it will be satisfactory for the pupil to be bounded at one point by a chord of said pupil, provided the reduction in area of the exit pupil does not exceed 2 percent. At the edge of the field of view, the minimum dimension of the exit pupil shall be not less than 45 percent of the diameter of the central exit pupil.

3.11.4 The true field of view of each of the two telescopic systems shall be not less than 7 degrees, 10 minutes.

3.11.5 The eye distance shall be not less than 14.0 mm.

3.11.6 The curvature of field, defined as the algebraic mean of the diopter scale readings when the eyepiece is focused successively on tangential and radial lines, shall not exceed two diopters at any point in the field of view. At 3 degrees of true field from the center of the field, the astigmatic difference shall not exceed four diopters and, at all angles, this difference shall be adjusted so as to give the best obtainable definition throughout the field, without sacrificing the best image quality and definition at the center.

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3.11.7 The limit of resolution within 1 degree of true field from the center of the field shall not exceed 4.0 seconds of arc (true field) for either vertical or horizontal lines at the same eyepiece focus.

3.11.8 The weight of the binoculars without strap or case shall not exceed 46 ounces.

3.11.9 The spraytight binoculars shall be in accordance with Drawings S2407-533198 and S2407-533370 and S2407-533370 through S2407-533380.

3.11.10 The watertight binoculars shall be in accordance with Drawings S2407-533198 and S2407-53315(through S2407-533164.

3.12 Type II - 6 power, 42 mm - aperture. -

3.12.1 The magnification of each of the two telescopic systems of the binocular shall be six diameters plus or minus 2 percent. The magnification of the two telescopes of a single binocular shall not differ by more than 1 percent.

3.12.2 The effective aperture of the objective shall be not less than 42.0 mm nor more than 44.0 mm.

3.12.3 The central exit pupil shall have a diameter not less than 42 mm divided by the measured magnification plus or minus 3 percent, except that it will be acceptable for the pupil to be bounded at one point by a chord of said pupil, provided the reduction in area of the exit pupil does not exceed 2 percent. At 5.75 degrees of true field from the center of the field the area of the exit pupil shall be not less than 40 percent of that of the central exit pupil.

3.12.4 The true field of view of each of the two telescopic systems shall be not less than 11 degrees, 45 minutes.

3.12.5 The eye distance shall be not less than 19.0 mm.

3.12.6 The curvature of field, defined as the algebraic mean of the diopter scale readings when the eyepiece is focused successively on tangential and radial lines, shall not exceed 1.0 diopter within 5.5 degrees of true field about the center of the field of view. At 4 degrees of true field from the center of the field the astigmatic difference shall not exceed 2.2 diopters and at all angles this difference shall be so adjusted as to give the best obtainable definition throughout the field, without sacrificing the best image quality and definition at the center.

3.12.7 The limit of resolution within 1 degree of true field from the center of the field shall not exceed 5.0 seconds of arc (true field) for either vertical or horizontal lines at the same eyepiece focus.

3.12.8 The weight of the binoculars without strap or case shall not exceed 59 ounces.

3.13 Workmanship. - The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Sampling. -

4.2.1 Inspection lot. - All binoculars of the same type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

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4.2.2 Sampling for group A tests.- A random sample of binoculars shall be selected in accordance with table I from each inspection lot by the Government inspector and shall be subjected to each of the group A tests specified in 4.3.1 with lot acceptance based on the sampling requirements of Standard MIL-STD-105.

Table I - Sampling for group A tests
AQL (approx.) = 1.5 percent defective.

Number of binoculars in inspection lot	Number of binoculars in sample	Number of binoculars nonconforming on any group A tests	
		Acceptance number	Rejection number
15 and under	10	0	1
16 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 and over	75	3	4

The Government inspector may institute tightened or reduced inspection in accordance with Standard MIL-STD-105.

4.2.3 Sampling for group B tests.- A random sample of binoculars shall be selected from each inspection lot by the Government inspector in accordance with table II and shall be subjected to each of the group B tests specified in 4.3.2.

Table II - Sampling for group B tests.

Number of binoculars in inspection lot	Number of binoculars in sample	Number of binoculars nonconforming on any group B test	
		Acceptance number	Rejection number
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	1	2
301 to 500	25	1	2
501 and over	35	2	3

4.2.4 Sampling for group C tests.- A sample number of binoculars in accordance with table III shall be selected by the Government inspector from the total number of binoculars in the contract and shall be subjected to the group C tests specified in 4.3.3.

Table III - Sampling for group C tests.

Number of binoculars on contract	Number of binoculars for group C tests
110 and under	2
111 to 500	3
501 and over	5

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4.3 Lot inspection and test. -

4.3.1 Group A tests. - Each of the sample binoculars selected in accordance with table I shall be subjected to each of the tests specified hereinafter which shall be conducted or witnessed by the Government inspector, and the results of each test compared with this specification. Failure to conform to this specification for any group A test shall be counted as a defect and the binocular shall be rejected. If the number of such nonconforming binoculars in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection tests provided all binoculars in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

Inspection and tests	References
Visual and dimensional inspection	3.2, 3.3.1, 3.3.2, 3.3.3, 3.5, 3.9
Definition test	3.3.4, 4.4.1
Collimation test	3.3.5, 4.4.2
Hinge test	3.5.3, 4.4.3
Shock test	3.7, 4.4.6
Resolution test	3.11.7, 3.12.7, 4.4.7

4.3.2 Group B tests. - Each of the sample binoculars selected in accordance with table II shall be subjected to each of the tests specified hereinafter which shall be conducted or witnessed by the Government inspector, and the results of each test compared with this specification. Failure to conform to this specification for any group B test shall be counted as a defect and the binocular shall be rejected. If the number of such nonconforming binoculars in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection tests provided all binoculars in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

Group B tests	Reference
Curvature of field	3.11.6, 3.12.6

4.3.3 Group C tests. - Each of the sample binoculars selected in accordance with table III shall be subjected to each of the tests specified hereinafter and the results of each test compared with this specification. In the event of any failure to conform to this specification for any group C test the contractor shall correct the cause of failure on all binoculars in the contract or replace the deficient binoculars with binoculars meeting the specification requirements.

Group C tests	Reference
Moisture resistance	3.6, 4.4.4, 4.4.5
Weight	3.11.8, 3.12.8

4.4 Test procedures. -

4.4.1 Definition. - The test objects shall be as follows:

(a) A round, brightly illuminated artificial star having an angular diameter not less than 10 nor more than 18 seconds of arc.
(b) A test target having black characters on a white background suitable for the detection of such defects as spherical and chromatic aberrations, astigmatism, curvature of field, and coma illuminated to a level of at least 100 foot candles.
(c) Natural targets.

4.4.2 Collimation. - A standard type collimator shall be used to determine conformance with 3.3.4.

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4.4.3 Hinge test. - At a temperature of 70° ± 5° Fahrenheit (F.), hinges shall support a torque force between 9-inch pounds minimum and 15-inch pounds maximum. The tests shall be made by applying weights at an arc of 5-inch radius, the center of which is the center of the binocular hinge. The weights applied to create the specified torque shall consist of the unsupported side of the binocular plus the additional weights as required acting on an arc of 5-inch radius.

4.4.4 Spraytight. - Prior to the test, the binoculars shall be stored in the testing room for at least 1 hour to permit the temperature of the binoculars to approximate room temperature (about 70°F.). The binoculars shall then be exposed to a shower of water at room temperature simulating a rainfall of moderate intensity for a period of 5 minutes in each of two positions - vertical with objective end up and vertical with eyepiece end up.

4.4.5 Watertight. - Prior to the test, the binoculars shall be stored in the testing room for at least 1 hour to permit the temperature of the binoculars to approximate room temperature (about 70°F.). The binoculars shall be tested by placing them in a water tank, in which the water is within 5 degrees of room temperature, objective end down for 5 minutes. The top surface of the binocular shall be at least 12 inches below the surface of the water.

4.4.6 Shock test. -

4.4.6.1 Binoculars shall be held with the eyepieces upward and dropped a distance of 6 feet into fine white sand 6 inches deep covered with a thin cloth.

4.4.6.2 A strong cord, practically free from stretch, such as sash cord, and of sufficient length, shall be tied around either the hinge pin or sleeve, or one of the hinge lugs and the binoculars dropped from a height of 6 feet without striking any object, so that the fall will be arrested by the cord.

4.4.7 Resolution. - The test object for resolution shall be a white chart ruled with black lines separated by white spaces the same width as the lines. It shall be illuminated to a level of at least 100 foot candles. The test shall be made with the aid of an auxilliary telescope of 6 power placed over the eyepiece and centered on its axis.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Preservation and packaging for domestic shipment - immediate use. - Preservation and packaging shall be in accordance with manufacturer's commercial practice.

5.1.2 For domestic shipment and storage and overseas shipment. -

5.1.2.1 Unless otherwise specified in the contract or order, the binoculars and accessories shall be cleaned by process C-1, and packaged in accordance with method II of Specification MIL-P-116.

5.1.2.2 The instrument and accessories shall be securely held within the carrying case by means of checks, straps, or other suitable means. In order to prevent shifting of binoculars and accessories during transit, cellulosic cushioning material conforming to Specification UU-C-843 shall be used. The carrying case shall be further wrapped in a minimum of 30 pound basis weight glazed kraft paper, with the glazed surface placed next to the surface of the carrying case and suitably secured with 2-inch kraft gummed tape, 60 pound basis weight. The binoculars and accessories shall then be placed in a fiberboard box conforming to symbol W6s or W6c of Specification JAN-P-108. Seams and closures shall conform to the applicable box specification.

5.2 Packing. -

5.2.1 For domestic shipment - immediate use. - The equipment preserved and packaged as specified in 5.1.2 shall be packed in accordance with the Consolidated Freight Classification Rules or Postal Regulations, whichever may be applicable. When fiberboard is used for container construction, the Mullen test shall be no less than 200 pounds. The gross weight of wood boxes shall not exceed 200 pounds.

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5.2.2 For domestic shipment and storage.- Unless otherwise specified in the contract or order, the equipment preserved and packaged as specified in 5.1.2 shall be packed in snug fitting wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes, conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631 or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape, or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds. Fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 For overseas shipment.- The equipment packaged as specified in 5.1.2 shall be packed in snug fitting wood cleated plywood, nailed wood or fiberboard boxes conforming to Specification JAN-P-105, style 2, 2-1/2 or 3 of Specification JAN-P-106, or symbol V3c or V3s of Specification JAN-P-108, respectively. Plywood shall conform to type A or B, condition I of Specification JAN-P-139. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to class A or B of Specification QQ-S-781. The gross weight of wood boxes shall not exceed 150 pounds, of fiberboard boxes, 70 pounds. Case liners will not be required.

5.2.4 For domestic and overseas shipment.- When practicable, each container shall contain an identical number of items and shall be of the same uniform size. In addition, all containers shall be new and shall be able to withstand storage, rehandling and reshipment without the necessity of repacking.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) If reticles are required (see 3.4).
(d) Whether spraytight or watertight binoculars are required (see 3.6).
(e) Carrying case required (see 3.8.2).
(f) Marking required (see 3.10).
(g) Whether domestic or overseas shipment is required; if domestic, the type required (see 5.1 and 5.2).

6.2 Standard samples of these binoculars may be seen upon application to the Navigational Equipment Section, Bureau of Ships, Navy Department, Washington 25, D.C.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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